SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer **ALL** questions in this section. Marks are indicated at the end of each question. Together they are worth 80% of the total marks for this examination.

Question 1 (21 marks – approximately 38 minutes)

Economics is essentially the study of the efficient allocation of limited resources. Scarcity implies that all decisions necessarily involve making trade-offs. One standard decision-making analysis used by economists is the so-called Marginal (cost-benefit) Analysis.

Required:

(a) (i) Explain how marginal analysis works.

(2 marks)

(ii) Define opportunity cost and its role in marginal analysis.

(4 marks)

(iii) Suppose a couple paid HK$2,500 for two pop music concert tickets. They subsequently lost the tickets and had to decide whether to pay another HK$2,500 to buy two new tickets. Assume that the couple is willing to pay at most HK$3,000 to watch the music concert.

(1) Explain how marginal analysis could be used to determine if the couple should buy two new tickets.

(2 marks)

(2) Explain how the SAME conclusion (as in part (1)) could be reached using "total" cost-benefit analysis.

(4 marks)

(b) (i) Define price elasticity of demand.

(2 marks)

(ii) Justify how price elasticity of demand can provide useful information to facilitate a firm's price decision-making in addition to that provided by demand and supply analysis.

(2 marks)

(iii) The Marvel Chocolate Shop ("MCS") estimates that at current prices the demand for its high-end chocolate has a price elasticity of –1.5 while the price elasticity for its low-end chocolate is –1.

If MCS raises the prices of high-end and low-end chocolates, interpret the impact on the sales and revenue of each product.

(5 marks)

<u>Question 2</u> (24 marks – approximately 43 minutes)

(a) (i) Define technical recession.

(2 marks)

(ii) Define business cycle.

(2 marks)

(iii) Describe the relationship between economic slowdown, recession and business cycle. [No definition for each individual term is required to be provided here.]

(2 marks)

(b) When the economy grows or slows down significantly, the government would typically engage in aggregate demand management through the use of monetary and fiscal policy.

(i) Define fiscal policy and explain how contractionary fiscal policy works.

(3 marks)

(ii) Define monetary policy and explain how expansionary monetary policy works.

(3 marks)

(iii) Explain how the central bank affects the money supply through the purchase of government bonds in the market. State the name of the policy tool.

(3 marks)

(iv) Describe TWO other monetary policy tools commonly used by central banks to conduct expansionary monetary policy.

(5 marks)

(c) Justify if the following statement is true or false:

Similar to all other central banks in developed economies, the Hong Kong Monetary Authority would react to an overheating economy by adopting a contractionary monetary policy.

(4 marks)

Question 3 **(17 marks – approximately 31 minutes)**

(a) Statistics has two broad meanings: one referring to data and the other to method. Statistical methods provide a powerful set of tools that allow researchers to analyse data and draw conclusions about the population studied.

Required:

State and describe the two major branches of statistics.

(2 marks)

(b) The following frequency distribution shows the monthly salaries of a sample group of employees in a logistics company.

Monthly salary (HK$'000)	No. of employees (f)	Mid point (x)	f.x	$f.x^2$
11 – 15	19			
16 – 20	32			
21 – 25	33			
26 – 30	28			
31 – 35	27			
36 – 40	17			
41 – 45	14			
Total	170			

Required:

(i) **Compute the missing figures in the table above.**

(3 marks)

(ii) **Calculate the mean and median of the monthly salary of the employees. Show supporting calculations.**

(4 marks)

(iii) **Calculate the standard deviation of the monthly salary of the employees. Show supporting calculations.**

(3 marks)

(iv) **Suppose there is an error in recording the monthly salary of the highest paid employee in the last class "41 – 45". The correct monthly salary of the employee should be HK$36,000. Analyse how the correct value affects the following measures.**

(1) **Mode**
(2) **Median**
(3) **Standard deviation**

(5 marks)

<u>Question 4</u> **(10 marks – approximately 18 minutes)**

(a) (i) Define population and sample in statistics.

(2 marks)

(ii) Identify TWO reasons why researchers usually work with samples instead of population.

(2 marks)

(b) Define sampling error and state TWO conditions under which the exposure to sampling error can be avoided.

(3 marks)

(c) Define random sampling. State and explain any TWO other methods of probability sampling.

(3 marks)

<u>Question 5</u> **(8 marks – approximately 14 minutes)**

(a) Contrast the difference between point estimate of the population parameter and confidence interval.

(4 marks)

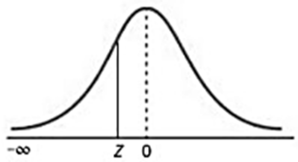
(b) An asset management company has conducted a survey to determine the proportion of its employees who took a professional examination in the past three years. The survey covered 300 of the employees, 230 of whom had taken the examination. Compute a 95% confidence interval of the population proportion of employees who took the examination in the past three years.

(4 marks)

* * * END OF EXAMINATION PAPER * * *

APPENDIX

The Cumulative Standardised Normal Distribution



Entry represents area under the cummulative standardised normal distribution from −∞ to Z

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-3.9	0.00005	0.00005	0.00004	0.00004	0.00004	0.00004	0.00004	0.00004	0.00003	0.00003
-3.8	0.00007	0.00007	0.00007	0.00006	0.00006	0.00006	0.00006	0.00005	0.00005	0.00005
-3.7	0.00011	0.00010	0.00010	0.00010	0.00009	0.00009	0.00008	0.00008	0.00008	0.00008
-3.6	0.00016	0.00015	0.00015	0.00014	0.00014	0.00013	0.00013	0.00012	0.00012	0.00011
-3.5	0.00023	0.00022	0.00022	0.00021	0.00020	0.00019	0.00019	0.00018	0.00017	0.00017
-3.4	0.00034	0.00032	0.00031	0.00030	0.00029	0.00028	0.00027	0.00026	0.00025	0.00024
-3.3	0.00048	0.00047	0.00045	0.00043	0.00042	0.00040	0.00039	0.00038	0.00036	0.00035
-3.2	0.00069	0.00066	0.00064	0.00062	0.00060	0.00058	0.00056	0.00054	0.00052	0.00050
-3.1	0.00097	0.00094	0.00090	0.00087	0.00084	0.00082	0.00079	0.00076	0.00074	0.00071
-3.0	0.00135	0.00131	0.00126	0.00122	0.00118	0.00114	0.00111	0.00107	0.00103	0.00100
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2388	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2482	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641